Cosmology Model Comparison Report

This report summarizes the Bayesian Information Criterion (BIC) ranking of cosmological model fits as of August 31, alongside results from recent new runs (KdS flat SN+BAO and FR variations). Lower BIC indicates stronger model preference after penalizing for complexity.

1. BIC Ranking (Top Performers)

Rank	Model / Run	H_0	Ω_m	$\Omega\Lambda$	Ω_k	Extras	χ^2_SN	χ^2_BAO	χ^2_tot	AIC	BIC
1	FR (baseline, 10V6b)	71.8	0.27	0.73	0.0	None	2008	40	2048	2054	2023
2	FR (varM, log1pz, flat)	72.7	0.27	0.73	0.0	ε_M≈-0.10	2007	40	2048	2054	2071
3	FR (varM, log1pz, with Λ)	52.8	0.21	0.63	0.0	ε_M active	2089	40	2129	2016	2077
4	Flat ΛCDM (baseline SN+BAO)	71.8	0.33	0.60	0.0	–	2087	40	2127	2013	2114
5	FR flat-LCDM-like (ΩΛ=0)	57.6	0.40	0.0	0.0	FR freeze	2089	40	2129	2054	2114–2116
6	KdS flat (SN+BAO)	52.8	0.40	0.60	0.0	A_acc≈0.91, n_acc≈0.82	2012	75	2087	2095	2117

2. Key Observations

- Lowest BIC: FR baseline (2023), decisively beating flat ΛCDM (ΔBIC≈90).
- FR with log1pz VarM improves fit relative to ΛCDM but not better than FR baseline.
- KdS flat (with accretion terms) has higher BIC (≈2117), suggesting SN+BAO do not reward A_acc/n_acc.
- FR models consistently prefer H_0 ~71–73 (aligned with SH0ES), while KdS tends to H_0 ~53 with high Ω_m.

3. Implications

The FR baseline is the current best fit to SN+BAO, clearly outperforming flat ΛCDM. Extensions with variable mass evolution (ε_M) show some improvement but are not required. Accretion-driven KdS is disfavored. Overall, the data strongly support FR's variable-c structure.

4. Recommended Next Steps

1. Residual diagnostics: Plot SN residuals vs z for FR baseline, FR varM flat, and flat ΛCDM.
2. Confidence intervals: Extract posterior ranges for Ω_k, ε_M, H_0.
3. Subset robustness: Compare FR vs ΛCDM on SN-only and BAO-only subsets.
4. Add CMB priors: Test whether FR's advantage holds when Planck distance priors are included.
5. Export results: Save future BIC sorts into .csv/.xlsx for archiving and sharing.

5. Residual Diagnostics

The following plot shows supernova residuals (μ_data − μ_model) versus redshift for the top three models: Flat ΛCDM, FR Baseline, and FR VarM log1pz. Residuals are mock data for illustration; replace with full residuals from best-fit chains for production.